

Each person in the world has visited Libya or there exists some other country which the person has not visited.

c) No one has climbed every mountain in the Himalayas.

$C(x, y)$: Person x has climbed mountain y of the Himalayas

Domain of x : Set of all people in the world

Domain of y : Set of all mountains in the Himalayas

$$\forall x \exists y \neg C(x, y) \quad \neg \forall x \exists y \neg C(x, y) \equiv \exists x \neg \exists y \neg C(x, y)$$

$$\text{(De Morgan's law of quantifiers)} \equiv \exists x \forall y \neg (\neg C(x, y)) \text{ (De Morgan's law of quantifiers)} \equiv \exists x \forall y C(x, y) \text{ (Double negation law)}$$

Some person in the world has climbed every mountain in the Himalayas.

d) Every movie actor has either been in a movie with Kevin Bacon or has been ~~in~~ in a movie with someone who has been in a movie with Kevin Bacon.

$A(x, y, z)$: Person x has acted with person y in movie z .

Domain of x : Set of all movie actors

Domain of y : Set of all movie actors

Domain of z : Set of all movies ^{Kevin Bacon}

$$\begin{aligned} & \forall x \forall y (A(x, \text{Kevin Bacon}, y) \vee \exists z (A(x, y, z) \wedge \exists w A(y, \text{Kevin Bacon}, w))) \\ & \neg \forall x (\exists y A(x, \text{Kevin Bacon}, y) \vee \exists y (\exists z A(x, y, z) \wedge \exists w A(y, \text{Kevin Bacon}, w))) \\ & \equiv \exists x \neg (\exists y A(x, \text{Kevin Bacon}, y) \vee \exists y (\exists z A(x, y, z) \wedge \exists w A(y, \text{Kevin Bacon}, w))) \\ & \text{(De Morgan's law of quantifiers)} \\ & \equiv \exists x (\neg \exists y A(x, \text{Kevin Bacon}, y) \wedge \neg \exists y (\exists z A(x, y, z) \wedge \exists w A(y, \text{Kevin Bacon}, w))) \\ & \text{(2nd De Morgan's Law)} \\ & \equiv \exists x (\forall y \neg A(x, \text{Kevin Bacon}, y) \wedge \forall y \neg (\exists z A(x, y, z) \wedge \exists w A(y, \text{Kevin Bacon}, w))) \\ & \text{(De Morgan's law of quantifiers)} \end{aligned}$$